

A BUSINESS ANALYSIS CASE STUDY

Healthcare Digital Platform

Business Analysis & Solution Design

Provider Lifecycle | Patient Care | Digital Healthcare Operations

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Executive Summary

The Challenge & Solution Approach

A healthcare digital platform faced operational challenges across provider onboarding, verification, appointment management, consultation, payment, notifications, and operational support — driven by manual processes, fragmented workflows, multiple handoffs, and limited visibility.

The proposed solution standardizes these processes through:

- Digital workflows & automated provider verification
- Configurable business rules & exception handling
- Integrated services, automated notifications, SLA monitoring
- Operational dashboards and KPI reporting

Objectives

- Digitize and standardize provider onboarding
- Automate credential and regulatory verification
- Improve approval, compliance and activation turnaround
- Standardize appointment and consultation workflows
- Integrate payment and notification workflows
- Enable real-time operational visibility

Key Deliverables

Business Requirements	Functional Requirements	Process Design	Solution Architecture
ERD & Data Dictionary	API Specifications	Agile User Stories & UAT	SQL Analytics & Tableau Dashboard

Business Context

INDUSTRY

HealthTech / Digital Healthcare Platform

A digital healthcare platform connecting healthcare providers and patients through digital healthcare services.

Existing Challenges

- Manual provider verification
- Multiple operational handoffs
- Fragmented workflows
- Limited process visibility
- Delayed provider activation
- Inconsistent SLA adherence
- Manual operational effort
- Limited centralized reporting

Transformation Need

- Faster provider onboarding
- Standardized workflows
- Reduced manual dependency
- Better regulatory compliance
- Real-time operational visibility
- Scalable digital healthcare operations

Business Problem Statement

Provider Onboarding

- Manual credential verification
- Multiple operational handoffs
- Long onboarding turnaround times
- Limited visibility into pending cases
- High operational effort
- Delayed provider activation

Healthcare Operations

- Fragmented appointment & consultation workflows
- Payment and notification dependencies
- Limited end-to-end operational visibility

Operational Challenges

- Inconsistent SLA adherence
- Limited centralized reporting
- Manual exception handling
- Dependency on Operations & Compliance teams

Business Impact

- Reduced operational productivity
- Delayed provider activation
- Poorer provider/patient experience
- Higher operational dependency
- Limited scalability

Objectives & Success Metrics

Strategic Objectives

- Digitize and standardize provider registration and onboarding
- Automate credential and regulatory verification
- Improve provider approval, compliance and activation turnaround
- Standardize appointment and consultation workflows
- Integrate payment and notification workflows
- Enable real-time operational visibility
- Improve auditability and compliance
- Reduce manual operational effort

Success Metrics

KPI	Baseline	Target
Average Onboarding Time	5 Business Days	< 24 Hours
Provider Approval Rate	60% (illustrative dataset)	> 80%
Manual Verification Effort	High	Reduced via automation
SLA Compliance	70% (Assumed Business Baseline)	> 95%
Dashboard Availability	Not Available	Real-Time Visibility

Stakeholder Analysis

Stakeholder	Key Responsibility	Expectation
Patients	Book appointments, consult providers, make payments	<i>Seamless booking</i>
Providers	Register, complete onboarding, deliver consultations	<i>Faster onboarding</i>
Operations	Verify documents, manage queue, resolve exceptions	<i>Reduced manual effort</i>
Compliance	Validate licenses, regulatory oversight	<i>Accurate verification & auditability</i>
Support	Resolve issues and escalations	<i>Faster resolution</i>
Product	Requirements & prioritization	<i>Improved business outcomes</i>
Engineering	Integrations & platform features	<i>Clear functional requirements</i>

NOTE QA is included separately as a delivery / support stakeholder within the SDLC scope, alongside the primary business stakeholders listed above.

Requirement Elicitation

Technique	Stakeholders	Output
Workshops	Product, Operations	Business Requirements
Interviews	Providers, Operations	User Needs
Process Observation	Operations	AS-IS Process
Data Analysis	BI / Operations	KPI Baseline
Document Analysis	SOPs, Policies	Business Rules & Gaps

Key Findings

- Multiple manual approvals
- Duplicate verification
- Poor SLA visibility
- High operational dependency
- No centralized reporting
- Delayed provider activation

Elicitation Approach

A blended technique-set combining stakeholder workshops, direct interviews, on-floor process observation and existing SOP/data analysis produced a triangulated baseline for business requirements, AS-IS gaps and KPI targets used across the rest of this analysis.

AS-IS Process



Challenges in the AS-IS Process



Manual verification



Multiple handoffs



No automated validation



Manual communication



Lack of real-time tracking



Limited dashboards



SLA breaches

Pain Point Analysis

Area	Existing Challenge	Business Impact	Opportunity
Provider Verification	Manual credential validation	Delayed onboarding	Automated verification
Appointment Management	Fragmented / manual workflow	Booking delays	Standardized appointment workflow
Consultation	Fragmented process	Operational dependency	Structured consultation workflow
Communication	Manual status updates	Limited transparency	Automated notifications
Reporting	Limited centralized reporting	Delayed decisions	Operational dashboards
SLA Monitoring	Reactive tracking	SLA breaches	Real-time SLA monitoring

Future State / TO-BE

Provider Onboarding (Automated)

01 Provider Registration	02 Document Upload	03 Mandatory Field Validation
04 Provider Verification API	05 Business Rules Evaluation	06 Auto Approval / Exception Routing
07 Automated Notification	08 Provider Activation	09 Dashboard Update in Real Time

Downstream Healthcare Journey

- Patient Management
- Appointment Management
- Consultation
- Payment Processing
- Notifications
- Support

Expected Improvements

Current	→	Future
Manual verification	→	Automated verification
Manual routing	→	Rule-based routing
Manual communication	→	Automated notifications
Static / limited reporting	→	Operational dashboards
Longer turnaround	→	Reduced onboarding time
Limited visibility	→	Real-time monitoring

Functional Solution Overview

Provider Lifecycle

- Digital Provider Registration
- Credential & Document Submission
- Automated Verification
- Provider Activation

Patient & Healthcare Journey

- Provider Search & Selection
- Appointment Management
- Consultation
- Secure Payment

Operations Management

- Verification Queue
- SLA Monitoring
- Exception Handling
- Compliance Review
- Support Ticket Management

Business Intelligence

- KPI Monitoring
- Executive Dashboards
- Operational Reporting

Functional Modules

Provider Onboarding

- Registration
- Verification
- Activation

Patient Management

- Patient profiles

Appointment Management

- Booking
- Appointment lifecycle

Consultation Management

- Consultation records
- Prescription / consultation completion

Payment Management

- Payment
- Refund

Notification Service

- Email
- SMS
- WhatsApp

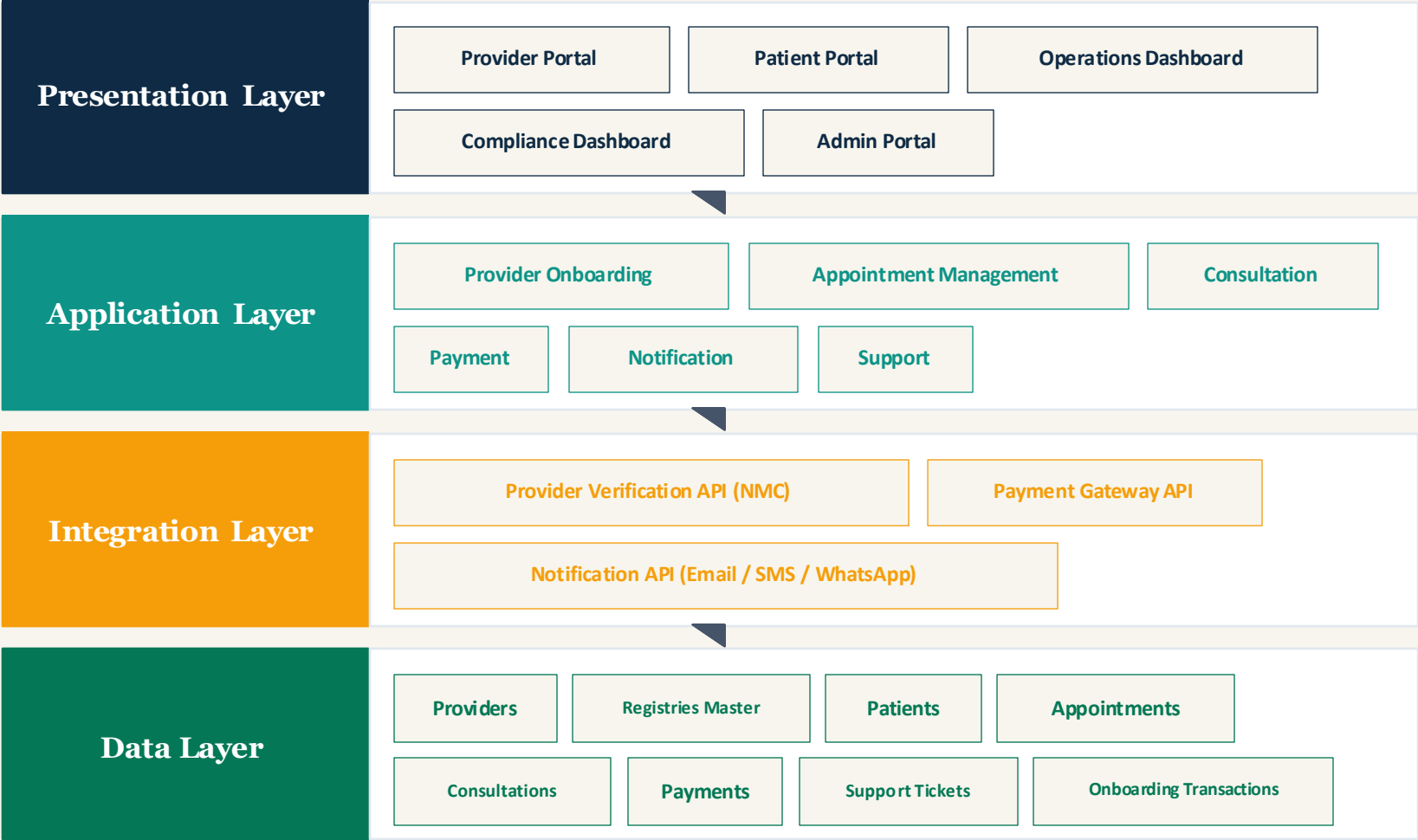
Support & Ticketing

- Support tickets
- Escalations

Analytics & Reporting

- Dashboard
- KPIs
- Reports

High-Level Solution Architecture



Technical Highlights

- Workflow-driven architecture
- API-enabled integrations
- Modular design
- Relational data structure
- Dashboard-ready analytics

Functional Requirements (FR-01 to FR-17)

Provider Registration

FR-01

 System shall allow providers to register using personal, professional and contact information.

FR-02

 System shall validate all mandatory fields before submission.

FR-03

 System shall generate a unique Provider ID upon successful registration.

Appointment Activation

FR-08

 System shall activate provider appointment scheduling after successful onboarding approval.

FR-09

 System shall prevent appointment booking for inactive providers.

Dashboard & Reporting

FR-12

 System shall display onboarding KPIs in an executive dashboard.

FR-13

 System shall track Approval Rate, Rejection Rate, Pending Queue, Activation Rate and SLA Compliance.

Provider Verification

FR-04

 System shall automatically invoke the Provider Verification API upon profile submission.

FR-05

 System shall validate registration details against the National Medical Council registry.

FR-06

 System shall automatically approve providers with an Active license status.

FR-07

 System shall route Suspended or Expired licenses to the Compliance queue for manual review.

Notifications

FR-10

 System shall send onboarding status notifications through Email, SMS or WhatsApp.

FR-11

 System shall notify Operations for pending verification beyond SLA thresholds.

Payment & Consultation

FR-14

 System shall initiate payment processing for booked consultations.

FR-15

 System shall update payment status based on gateway response.

FR-16

 System shall create consultation records after successful appointment completion.

Audit Trail

FR-17

 System shall record onboarding status changes and relevant user actions in an audit trail for authorized review.

Non-Functional Requirements

Category	Requirement
Performance	Verification responses should be processed within defined SLA limits where external services are available.
Availability	Platform should be available 24×7, excluding scheduled maintenance.
Scalability	Support increasing provider onboarding volumes without significant performance degradation.
Reliability	Failed API requests should be retried and logged.
Security	Sensitive provider and patient data encrypted in transit and at rest.
Auditability	All onboarding lifecycle events recorded.
Usability	Workflow should be intuitive and minimize manual effort.
Maintainability	Business rules configurable with minimal development effort.

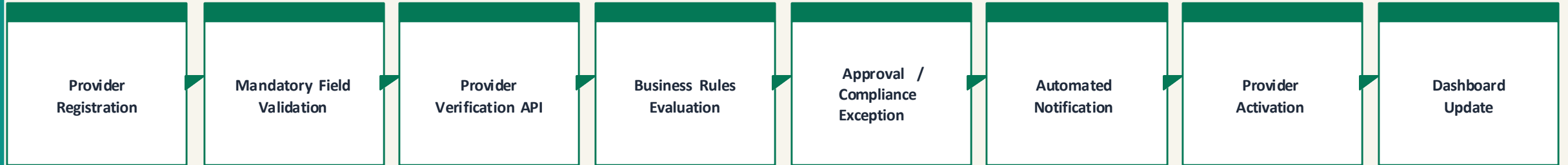
AS-IS Process Flow



5 Business Days

Average Provider Onboarding Time (Baseline)

TO-BE Process Flow



< 24 Hours

Target Average Onboarding Time

Business Rules Engine

BR-Rule-01

A provider cannot be activated without successful regulatory verification.

BR-Rule-02

Providers with Suspended or Expired licenses must be routed to the Compliance queue.

BR-Rule-03

Appointment booking is permitted only by Active providers.

BR-Rule-04

Payment must be successfully completed before appointment confirmation.

BR-Rule-05

Pending verification cases exceeding the defined SLA must be escalated to Operations.

BR-Rule-06

Every onboarding status change must generate an audit log entry.

BR-Rule-07

Only authorized roles can approve or reject provider applications.

BR-Rule-08

Dashboard metrics must refresh based on the latest onboarding transaction data.

Business Capability Map & Functional Architecture

Capability	Description	Key Functions
Provider Onboarding	Provider registration & activation	Registration, verification, activation
Patient Management	Patient lifecycle	Profiles
Appointment Management	Booking lifecycle	Booking, appointment management
Consultation Management	Consultation lifecycle	Consultation records, prescriptions
Payments	Transaction handling	Payment, refund
Notifications	Stakeholder communication	Email, SMS, WhatsApp
Support	Issue management	Support tickets, escalations
Analytics & Reporting	Operational intelligence	KPIs, dashboards, reports

Integration Layer

NMC Provider Verification API | Payment Gateway API | Notification API

Data Layer

Providers | Registries Master | Patients | Onboarding Transactions | Appointments
| Consultations | Payments | Support Tickets

Solution Impact

Expected KPI Improvements

KPI	Baseline	Target / Expected
Average Onboarding Time	5 Business Days	< 24 Hours
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NOTE Figures are **expected / target** outcomes for this portfolio case study — not measured production results.

Expected Business Benefits

- Reduced onboarding turnaround time
- Reduced manual verification effort
- Improved regulatory compliance
- Faster provider activation
- Improved provider experience
- Reduced support effort
- Improved operational visibility
- Better decision-making via dashboards & KPIs

Technical Deliverables & Project Assets

Deliverable	Purpose
Business Requirements Document	Defines scope, objectives, assumptions and business requirements
Functional Requirements	Documents system behaviour and validation logic
Non-Functional Requirements	Defines quality and operational expectations
Business Rules	Defines decision and validation logic
Functional Architecture	Defines platform layers and components
ERD & Data Dictionary	Defines data entities, relationships and fields
API Contracts	Provider Verification, Payment Gateway, Notification APIs
Agile Backlog	16 User Stories
Gherkin Acceptance Criteria	Defines testable acceptance conditions
UAT Test Pack	Business validation scenarios
SQL Analytics	KPI and operational analytics
Tableau Dashboard	Operations Dashboard
Governance & Compliance	Audit, RBAC and regulatory considerations

Tools Used

- Microsoft Word
- Microsoft PowerPoint
- PostgreSQL
- PostgreSQL SQL
- Tableau Public

16

User Stories

Agile backlog scope

Project Summary & Business Impact

Business Problem Solved

Designed a Healthcare Digital Platform to standardize provider onboarding, automate provider verification, improve compliance, reduce onboarding delays, and enhance operational visibility through workflow automation, integrations and analytics.

Key Deliverables

- End-to-End Business Analysis
- AS-IS / TO-BE Process Design
- Stakeholder Analysis
- Business & Functional Requirements
- Business Rules
- ERD & Data Dictionary
- API Integration Design
- Agile User Stories & UAT
- SQL Analytics
- Operations Dashboard
- Governance & Compliance

Core Competencies

- Business Analysis
- Requirements Engineering
- Process Mapping
- Functional Design
- API Analysis
- Data Modeling
- SQL Analytics
- Dashboard Design
- User Story Writing
- UAT Planning
- Governance & Compliance
- Stakeholder Management

Why This Project Matters

- Demonstrates end-to-end BA lifecycle
- Combines healthcare domain with technical analysis
- Demonstrates requirements engineering & functional design
- Demonstrates data-driven decision making
- Demonstrates cross-functional stakeholder management